HW 3 - Zeshui Song

4.6

6. Calculate the power dissipated in the 1 Ω resistor of Fig. 4.36.

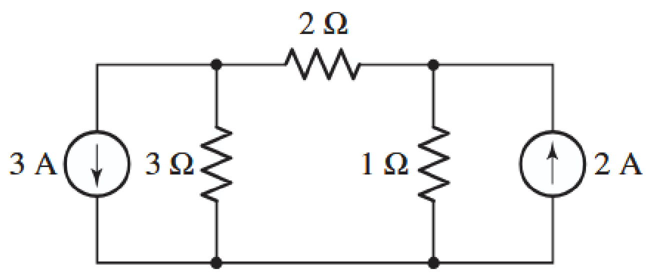

FIGURE 4.36

4.12

12. Using the bottom node as reference, determine the voltage across the 5 Ω resistor in the circuit of Fig. 4.42, and calculate the power dissipated by the 7 Ω resistor.

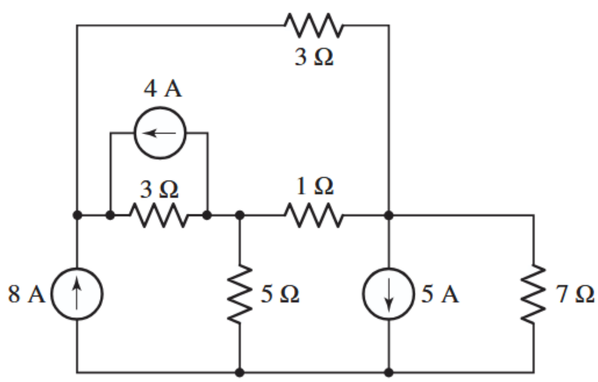

FIGURE 4.42

4.20 20. For the circuit of Fig. 4.50, determine all four nodal voltages.

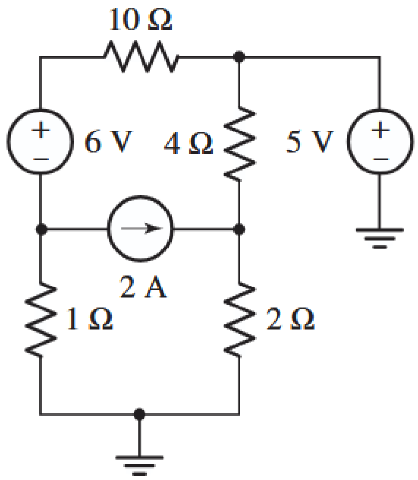

FIGURE 4.50

4.32 32. Determine the currents flowing out of the positive terminal of each voltage source in the circuit of Fig. 4.60.

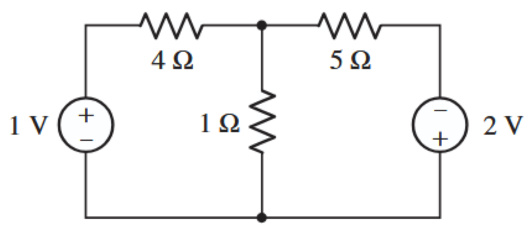

FIGURE 4.60

4.36 36. Calculate the power dissipated by each resistor in the circuit of Fig. 4.63.

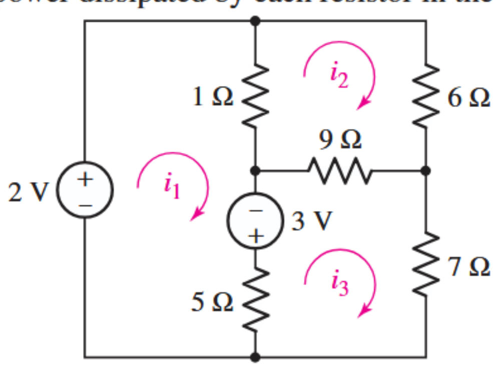

FIGURE 4.63

4.46 46. For the circuit of Fig. 4.73, determine the mesh current i_1 and the power dissipated by the 1 Ω resistor.

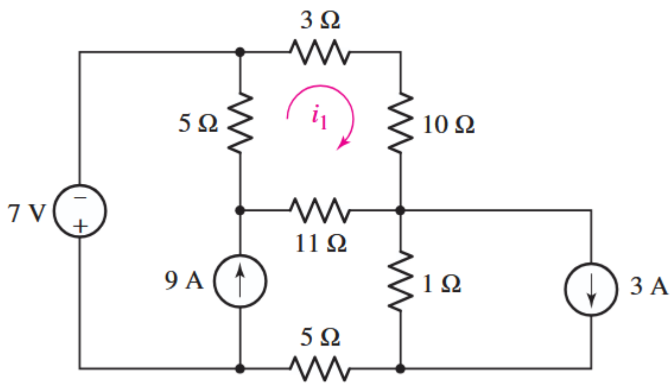

FIGURE 4.73

4.56 56. Solve for the voltage v_x as labeled in the circuit of Fig. 4.82 using (*a*) mesh analysis. (*b*) Repeat using nodal analysis. (*c*) Which approach was easier, and why?

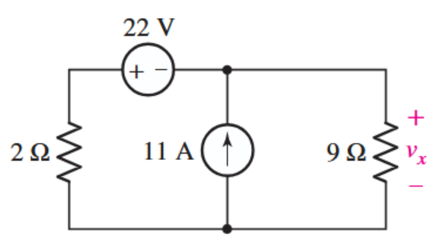

FIGURE 4.82

4.57 57. Consider the five-source circuit of Fig. 4.83. Determine the total number of simultaneous equations that must be solved in order to determine v_1 using (*a*) nodal analysis; (*b*) mesh analysis. (*c*) Which method is preferred, and does it depend on which side of the 40 Ω resistor is chosen as the reference node? *Explain your answer.*

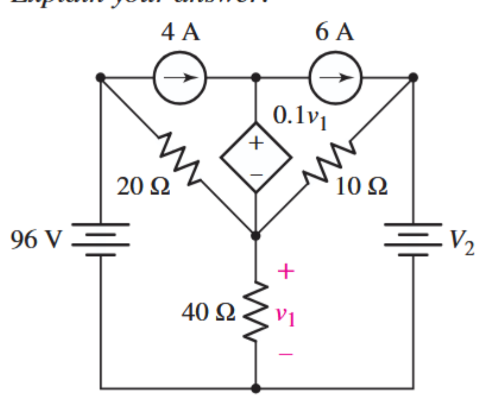

FIGURE 4.83

5.3

3. Considering the circuit of Fig. 5.48, employ superposition to determine the two components of i_8 arising from the action of the two independent sources, respectively.

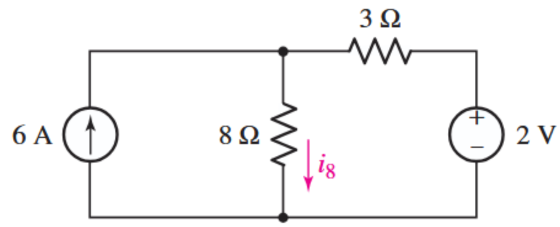

FIGURE 5.48

5.5

5. (*a*) Using superposition to consider each source one at a time, compute i_x. (*b*) Determine the percentage of i_x arising from each source. (c) Adjusting only the current source, alter the circuit to double i_x.

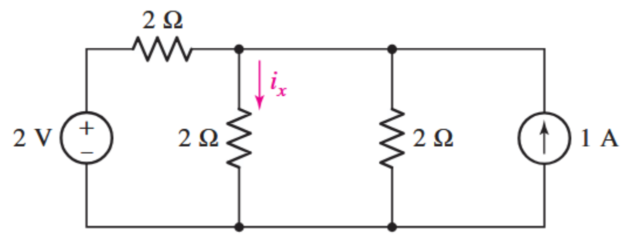

FIGURE 5.50

5.11

11. Employ superposition principles to obtain a value for the current I_x as labeled in Fig. 5.56.

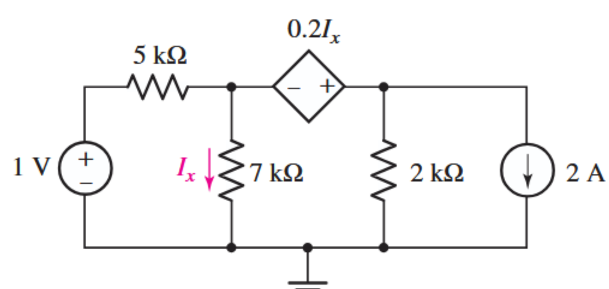

FIGURE 5.56

5.18 18. (*a*) Using repeated source transformations, reduce the circuit of Fig. 5.62 to a voltage source in series with a resistor, both of which are in series with the 6 MΩ resistor. (*b*) Calculate the power dissipated by the 6 MΩ resistor using your simplified circuit.

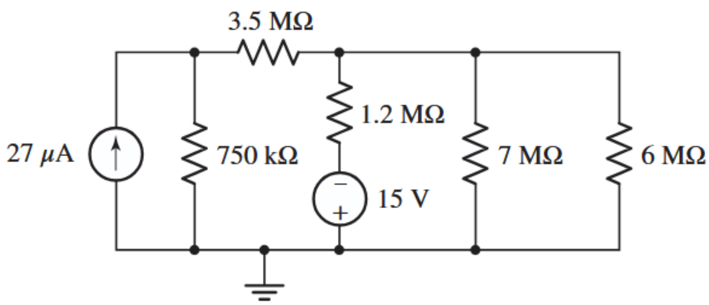

FIGURE 5.62